Does Situational AI Represent an Emerging Programming Paradigm?

Situations as the Core Abstraction for Reasoning, Decision-Making, and Execution

Introduction

For most of the past 70 years, software has been structured with the assumption that relevant behaviors of a system can be determined before execution starts. Procedural programming encodes that assumption in ordered sequences of instructions and explicit control flow. Object oriented programming re-packages it in the form of class hierarchies and encapsulation. Both approaches are powerful as they are explicit, yet both depend on the assumption that the designer can predict beforehand what the system will have to do (Van Roy, 2009). If tasks are stable and contexts predictable this premise works fine. It fails when systems are required to operate in environments that change constantly, where what is important changes faster than rules can be changed. Machine learning relaxed that rule by letting systems learn patterns from data rather than hand-coded logic, but the loosening brought a different kind of rigidity with it. ML systems rely on training pipelines, versioning, lifecycle management, and continuous monitoring. These systems learn within a distribution but do not update their understanding of the context in which their outputs are used (Sawant et al., 2021). Agentic and AI-native systems pushed the frontier by adding tool use, task decomposition, and multi-step orchestration into single operational frameworks. However, the literature on these systems always describes their architecture in terms of workflows, execution loops and governance layers, rather than a persistent representation at the level of the situation of what is going on (A Survey on Agent Workflow, 2025; A Survey of Workflow Optimization for LLM Agents, 2026). All generations of software design have been getting closer to adaptive behavior but none have placed the situation itself at the heart of the computational architecture. This paper answers a simple question: is it true that Situational AI is a new programming paradigm that uses situations as the basic abstraction for reasoning, decision-making and execution, rather than prompts, tasks or workflows? It is not a purely definitional question. If situations can be the fundamental unit of computation—structured objects that include context signals, temporal state, role and responsibility, risk assessment, expected response plans, and escalation logic—then the implications for software design, autonomous systems, and human-AI collaboration go well beyond improvements in prompting or orchestration (Situational AI: Conceptual and Theoretical Foundations, n.d.). In this review, we leverage the literature in traditional programming paradigms, machine learning systems, agentic architectures, situational awareness research, and human-AI collaboration to evaluate that claim, bring to light the structural gap that underlies it, and reflect on both the evidence that supports it and the limitations that qualify it.

Traditional Programming Paradigms

Agentic and AI-Native Systems

(Machine Learning, 2020; Recent Years, Machine Learning Engineering Practices, 2025). Following this, Serban and Visser find that software engineering practices for ML are still maturing as a field, with much variation across teams in how they deal with testing, documentation and governance after deployment (Serban & Visser, 2021). The common thread across these results is a consistent pattern: the system is able to alter predictions within a learned distribution, but it does not automatically maintain a strong understanding of the context into which those predictions are being put.The pipeline is doing interpretive work that the system can't do yet. Research on ML-enabled systems corroborates this from different angles in a collaborative manner. Nahar et al. (2022) show that constructing and maintaining ML systems require significant communication, documentation, and coordination of model building, testing, deployment, and maintenance, meaning the system's behavior is embedded in a human-maintained lifecycle, not a self-contained situational model. The gap between what the system can infer and what a human must manage on its behalf is precisely the structural gap that Situational AI is designed to fill – not by replacing the ML layer but by adding a cognitive layer above it that can reason about the context in which that system is operating. Systems with Agency and AI-First

Agentic systems represent the most active form of current AI software design, combining planning, tool use, task decomposition, and multi-step execution into operational frameworks that far exceed what standard ML pipelines can accomplish within a single session. The 2025 AI Agent Index documents the rapid growth of these systems across sectors, noting that agent architectures are increasingly deployed in enterprise, scientific, and coordination-intensive settings (Staufer et al., 2025). The literature on agent workflows has correspondingly matured, with Yu and colleagues identifying planning, decomposition, collaboration, and execution loops as the central building blocks of current agent design—a finding that reflects both the genuine capability of these systems and the procedural logic that still organizes them (Yu et al., 2025).

That procedural logic persists even in the most sophisticated current architectures. Adimulam and colleagues describe multi-agent systems through the lens of architectures, protocols, and governance mechanisms, confirming that explicit control remains a defining feature of these designs even when agents can coordinate autonomously (Adimulam et al., 2026). Brodimas and colleagues extend this to infrastructure, showing how intent-based orchestration translates high-level goals into structured service actions in ways that improve reliability and accountability but still produce workflow-based intelligence (Brodimas et al., 2025). Wang and colleagues make the same point from an AI-native architecture perspective, framing current AI-native design as task-driven and system-level, with emphasis on observability and integration rather than on persistent situational reasoning (Wang et al., 2025).

The collaboration literature adds an important dimension by showing that even advanced agentic systems still depend heavily on human decisions at multiple stages. Nahar and colleagues show that collaboration challenges in ML-enabled systems persist even as orchestration capabilities improve, because uncertainty, interpretation, and responsibility continue to require human involvement when system outputs affect downstream decisions (Nahar et al., 2022). The path ahead for agentic AI, as surveyed in recent literature, identifies this dependency as one of the central challenges for the field: current systems can coordinate action well, but coordination is not yet the same as situation-centered reasoning (The Path Ahead for Agentic AI, 2026). The

agentic literature as a whole represents a major advance in what AI systems can do, while simultaneously confirming the structural gap that Situational AI proposes to fill.

Situational Awareness as Intellectual Background

Before defining Situational AI, it is necessary to situate the concept in relation to the situational awareness literature that provides its intellectual lineage. Situational awareness is most commonly described through Endsley's three-stage model: perception of environmental elements, comprehension of what those elements mean in context, and projection of future states based on current understanding (Endsley, 1995, as cited in Situational AI: Conceptual and Theoretical Foundations, n.d.). That framework originated in aviation psychology and has since been applied across human factors engineering, military operations, healthcare, and autonomous systems design. It provides a well-validated vocabulary for thinking about adaptive action in dynamic environments and has generated a substantial empirical literature on how awareness is built, maintained, and lost under operational conditions.

Work by Kaufman and colleagues illustrates how the model has been extended to joint human-autonomous action. Their research frames situational awareness as something shared and operational rather than individual or perceptual, showing that effective coordination between humans and autonomous systems requires both parties to maintain and update a common situational frame in real time (Kaufman et al., 2024). Research on situation awareness in human-AI teams extends this further, showing that effective awareness must be distributed across multiple agents and layers—taskwork, agent-level, and teamwork dimensions alike—and maintained dynamically rather than held in a single static representation (Nofal et al., 2023). Man and colleagues contribute a complementary finding by showing that situational awareness matters in 3D vision-language reasoning, where context shapes how systems interpret perceptual inputs—a result that underscores the relevance of awareness mechanisms beyond physical navigation (Man et al., 2024).

The broader situational awareness literature situates the concept across multiple disciplines. Foundational accounts describe situational awareness as a cognitive capacity enabling effective action in dynamic settings by integrating perception, comprehension, and anticipation of environmental states (Situational Awareness in Social Sciences and Humanities, n.d.). The prospective analysis in Situational Awareness: The Decade Ahead extends this framing to emerging technological and organizational environments, arguing that perception, anticipation, and adaptive response are the core functions that autonomous and collaborative systems must support (Situational Awareness: The Decade Ahead, 2024). That background is essential for understanding why context and timing matter in autonomous systems, but it does not by itself constitute the argument this paper makes. Situational awareness describes a cognitive or operational capability—how agents perceive and respond to their environment. Situational AI makes a stronger and more specific architectural claim: that situations should become the primary unit of computation itself, not merely a prerequisite for effective response.

The Current Gap in AI Systems

The review of traditional programming, ML-enabled systems, and agentic architectures converges on a consistent structural gap: current AI systems can process prompts, route tasks, and maintain limited context, but none of them treats situation as the primary unit of reasoning. That gap is not a matter of scale or performance. It is architectural. Even the most capable agentic systems described in the literature are organized around episodes, workflows, or training cycles rather than around a persistent situational layer that survives across prompts, tasks, and sessions.

Work by Nahar and colleagues makes this gap especially visible at the human-system interface. Their findings show that deployed ML systems remain embedded in human-maintained lifecycles—documentation, coordination, and process management are doing much of the interpretive work that the system cannot do for itself (Nahar et al., 2022). Yufik and Malhotra extend the argument philosophically, distinguishing genuine situational understanding from pattern completion: understanding, in their formulation, is what allows humans and machines to perform adequately in novel situations where prior experience is incomplete, and current AI systems achieve pattern completion without sustaining the kind of situational understanding that novelty demands (Yufik & Malhotra, 2021). Apple's empirical work on reasoning models points in the same direction: stronger reasoning traces do not guarantee reliable performance as task complexity increases, and beyond a threshold, reasoning quality degrades rather than deepens—suggesting that scaling existing architectures is unlikely to produce the durable, situation-centered reasoning that complex environments require (Apple Machine Learning Research, 2026).

The agentic literature corroborates this from an architectural perspective. Man and colleagues show that situational awareness matters in language and vision systems precisely because meaning depends on context, not on the isolated processing of inputs (Man et al., 2024). The path ahead for agentic AI remains constrained by the same structural feature: systems that coordinate steps, manage tools, and follow control logic are active in execution but episodic in cognition (The Path Ahead for Agentic AI, 2026). Adimulam and colleagues, Yu and colleagues, Brodimas and colleagues, and Wang and colleagues all describe systems that represent major advances in capability while sharing the same limitation: none of them maintains a durable, situation-centered representation that integrates context, continuity, and adaptive interpretation across the full arc of an interaction. Situational AI is proposed precisely to name that missing layer.

Situational AI as the New Abstraction

Situational AI rests on a specific architectural claim: that a situation, not a prompt or workflow, should serve as the primary unit of reasoning and execution. The Situation-Centric Behavior and Cognition Model (SCBCM) formalizes this claim by defining a situation as a structured tuple $S = (C, T, R, P, E, X)$, where C represents contextual signals, T the temporal state, R role and responsibility, P risk and priority assessment, E the expected response plan, and X escalation rules (Situational AI: Conceptual and Theoretical Foundations, n.d.). This formalization transforms situations from abstract concepts into computable objects that can be instantiated, stored, retrieved, and updated across the full lifecycle of an interaction rather than discarded after each prompt-response cycle.

The cognitive foundations for this claim are drawn from several converging research traditions. Endsley's situation awareness model maps directly onto SCBCM's layered architecture: the input layer aligns with perception, situation recognition with comprehension, and planning and execution with projection (Situational AI: Conceptual and Theoretical Foundations, n.d.). The Belief-Desire-Intention (BDI) framework, which models agents through beliefs about the world, desires as goals, and intentions as committed plans, is operationalized in SCBCM's situation tuple: contextual signals encode beliefs, role and priority capture desires, and response plans formalize intentions (Situational AI: Conceptual and Theoretical Foundations, n.d.). Zhang and colleagues provide architectural grounding by arguing for organizing AI around different layers of abstraction rather than flattening reasoning into a single mechanism, a principle that aligns directly with SCBCM's layered design (Zhang et al., 2019). A scenario-driven cognitive approach to next-generation AI memory reinforces the need for structured, long-term situation representation beyond what episodic prompt-response architectures can provide (A Scenario-Driven Cognitive Approach to Next-Generation AI Memory, 2025).

The limitations of large language model-based agents make the case for this abstraction concrete. LLMs operate on finite context windows without inherent long-term storage, treat inputs as episodic without tracking evolving situations over time, and lack built-in safeguards for high-risk scenarios (Situational AI: The Missing Cognitive Layer in AI Agents, n.d.). These are not incidental engineering shortcomings but architectural features: the prompt is the unit of processing, and everything outside the current context window is unavailable to the system. SCBCM addresses these gaps through a hybrid graph-vector memory layer that preserves structured situation representations across sessions, a TAO (Think-Act-Observe) loop that adapts Boyd's OODA cycle to AI decision-making by embedding belief-desire-intention mechanisms at the orientation stage, and an audit and explainability layer that makes situation transitions traceable and legible to human oversight (Situational AI: The Missing Cognitive Layer in AI Agents, n.d.; Situational AI: Conceptual and Theoretical Foundations, n.d.). The result is a system that is not merely responsive but situationally organized—one that reasons within situations, tracks them over time, and adapts when the situation changes.

Toward a Concrete System Model

The Situational AI paradigm gains engineering credibility when the architectural claim is grounded in a concrete operational model rather than left at the level of theory. Research on the instructional design of Situational AI coworker systems proposes a layered architecture comprising five functional components: a local language model for contextual interpretation, a situational knowledge base combining graph and vector representations, a continuous monitoring loop for detecting candidate situations from incoming data streams, an escalation engine for deciding between autonomous action, human confirmation, and deferred response, and a human feedback capture system for refining situational reasoning over time (Track 4: Instructional Guide for Applying Situational AI, n.d.). These components are not independent modules but interdependent parts of a single operational loop, each depending on the others to produce situation-centered behavior.

The explainability literature provides strong convergent support for this architecture. Sanneman and colleagues establish that a situation awareness-based framework for explainable AI should evaluate explainability in terms of what the system's reasoning reveals about its situational understanding, not simply in terms of output accuracy (Sanneman et al., 2020). Bansal and colleagues extend this by showing that transparency, explainability, and situation awareness function as integrated design requirements for human-AI teaming rather than separable features (Bansal et al., 2023). A conceptual framework for AI-based decision systems in critical contexts reinforces both findings by arguing that high-stakes systems require explicit logic for state assessment, response selection, and oversight rather than only reactive generation (A Conceptual Framework for AI-based Decision Systems in Critical Contexts, 2025). In other words, the audit and escalation functions are not optional additions to a Situational AI architecture; they are constitutive of what makes the system trustworthy enough to operate with bounded autonomy.

The cognitive architecture literature adds precision to the memory and reasoning layers. Work on cognitive interaction layers for neuro-symbolic AI suggests a design in which perception, symbolic structure, and reasoning are kept distinct rather than compressed into a single undifferentiated process, which allows the system to identify what is happening, represent it in a durable and structured form, and decide how to respond without losing the contextual frame (Cognitive Interaction Layers for Neuro-Symbolic AI, 2025). The human-like AI coworker literature aligns with this by treating cognition as more than response generation—the system must maintain continuity across situations, carry forward relevant history, and update its situational model as conditions change (Rosen, n.d.). Together, these sources establish that a concrete Situational AI architecture is not only theoretically coherent but practically realizable with existing computational components, provided those components are organized around situation as the primary abstraction.

Situational AI in Human-Like AI Coworker Systems

The practical value of Situational AI is most visible in systems designed to work alongside people rather than execute isolated tasks. A human-like AI coworker, as described in research on Situational AI design, is a system that continuously monitors digital environments, interprets situations as they emerge, decides autonomously within defined boundaries, and escalates to human oversight when confidence is insufficient or stakes are high (Track 4: Instructional Guide for Applying Situational AI, n.d.). This operational profile cannot be satisfied by prompt-responsive or workflow-capable systems alone; it requires a situation-centered architecture that maintains context, tracks change, and reasons about when and how to involve the human partner.

The human-AI teaming literature articulates the requirements that coworker systems must satisfy. A situation awareness framework for human-AI teaming argues that effective collaboration depends on both the human and the AI maintaining an accurate and current representation of what each party knows, what the task requires, and how the context is changing over time (A Situation Awareness Framework for Human-AI Teaming, 2023). Research on user cognitive state shows that awareness of the human partner's workload and attention can improve collaborative outcomes, reinforcing the need for a system that is sensitive to the person and the

situation together rather than only to the task in isolation (How Awareness of a User's Cognitive State Can Improve Human-AI Collaboration, 2024). Situational adaptive autonomy research treats the level of system autonomy as something that should vary dynamically with the situation rather than remain fixed, which corresponds directly to the escalation logic at the heart of SCBCM (Situational Adaptive Autonomy in Human-AI Cooperation, 2023).

The agentic architecture literature provides the engineering scaffolding for these requirements. A comprehensive survey of agentic AI architectures shows that reasoning, memory, tool use, and action can be organized into structured workflows that preserve context and track responsibilities across multi-step tasks (Agentic AI: A Comprehensive Survey, 2025). Research on multi-agent orchestration identifies reasoning, memory, communication protocols, and execution governance as the interdependent building blocks of capable agent systems, noting that the central challenge is organizing those components to support reliable, bounded autonomy rather than simply maximizing task throughput (Adimulam et al., 2026; The Path Ahead for Agentic AI, 2026). What distinguishes a Situational AI coworker from a capable agentic system is precisely what SCBCM adds: a persistent situation model that includes the human partner, the current task state, the history of the interaction, and the escalation path—making the system not only more capable but also more trustworthy and more legible to the people it works alongside.

Counterarguments and Limitations

A rigorous evaluation of the Situational AI paradigm claim requires engaging seriously with the counterarguments. Three substantive objections deserve treatment, and each is strengthened rather than weakened by considering the evidence carefully.

The first objection is that Situational AI may be better understood as a synthesis of existing human-AI teaming research rather than a genuinely new paradigm. Prior work already identifies responsiveness, situation awareness, and flexible decision-making as central requirements for effective human-AI collaboration in complex settings (Hagemann et al., 2023). The ATSA framework models human and AI situational awareness as a dynamic, bidirectional interaction, which means the relational structure that Situational AI proposes already has representation in the human factors literature (Wen et al., 2023). Work on taskwork SA, agent SA, and teamwork SA in human-AI teams demonstrates that multiple simultaneous layers of situational awareness matter for performance, further suggesting that Situational AI builds on an established research base rather than introducing wholly new problems (Nofal et al., 2023). The appropriate response to this objection is not to deny the intellectual lineage but to specify what SCBCM adds: a formal computational architecture in which situations function as first-class objects rather than as evaluation criteria or design guidelines. The novelty is architectural, not conceptual.

The second objection concerns the extent to which existing cognitive and architectural approaches already occupy the same design space. The team-centered AI literature addresses technical requirements such as cognitive competence, reinforcement learning, and semantic communication that overlap substantially with what a Situational AI architecture requires in practice (Hagemann et al., 2023). Research on AI and human cognition, including work on knowledge collapse under AI dependency, raises a related concern: if Situational AI systems

become the primary interface through which people manage complex work, the long-term effects on human situational competence are uncertain and potentially negative (AI, Human Cognition and Knowledge Collapse, 2026). These are legitimate concerns that a responsible development of the Situational AI paradigm must address directly, particularly in the design of escalation logic and human oversight mechanisms.

The third limitation concerns the maturity of the literature and the practical feasibility of the paradigm. Both the teaming literature and the governance literature leave open questions about how shared situational awareness should be maintained across human-AI boundaries, how information should be represented to support both machine reasoning and human comprehension, and how coordination should work when context changes rapidly (Hagemann et al., 2023; Wen et al., 2023). Manheim and Homewood make a particularly important governance point: oversight and control are not equivalent, and meaningful governance depends on whether a system's autonomy can be bounded in real time rather than only in principle (Manheim & Homewood, 2025). A Situational AI system is only as valuable as the reliability of its escalation mechanisms, and those mechanisms are still an open engineering problem in environments characterized by high uncertainty and fast-changing conditions. The strongest version of the Situational AI argument acknowledges these constraints while maintaining that the architectural framing it proposes represents a meaningful advance over existing approaches to AI system design.

Conclusion

The evidence reviewed across eleven sections of literature supports a substantive but qualified affirmative answer to the research question: Situational AI does represent an emerging programming paradigm in which situations, rather than prompts, tasks, or workflows, serve as the core abstraction for reasoning, decision-making, and execution. The claim is qualified because the paradigm is still taking shape and builds on established prior work in human factors, cognitive architecture, and agentic systems design. It is substantive because no existing approach fully resolves the gap between execution capability and persistent situational understanding, and because the Situation-Centric Behavior and Cognition Model provides a formal, computable architecture that gives the paradigm claim engineering rather than merely conceptual weight.

The trajectory of the literature makes the structural gap visible across multiple research traditions. Traditional programming paradigms assume predetermined behavior; ML systems improve adaptability but remain pipeline-dependent; agentic and AI-native systems expand execution capability but stay organized around orchestration rather than situational reasoning. Human-centric AI architecture research argues that future systems should be built around people and context rather than only around outputs (Human-Centric Artificial Intelligence Architecture for Industry 5.0, 2023), and the situational awareness framework for explainable AI demonstrates that system design depends fundamentally on whether the AI can help humans understand what is happening and why it is acting (Sanneman et al., 2020). A foundational architecture review for AI agents in healthcare reaches a similar conclusion, finding that current architectures need a stronger organizing layer to connect memory, reasoning, and action coherently (A Foundational Architecture for AI Agents in Healthcare, 2025). An emerging framework for AI agent architectures in enterprise settings reinforces the practical urgency of

this need by showing that production agentic systems require structured reasoning, memory, and oversight capabilities that current frameworks still do not provide by default (AI Agents: Evolution, Architecture, and Real-World Applications, 2025).

What unites these findings is a recognition that the next meaningful step in AI design is neither better prompting nor more sophisticated workflow management, but an architecture that holds context over time, updates that context as conditions change, and reasons from the full situation rather than from the immediate instruction. Situational AI proposes that architecture by making situation the primary unit of intelligence. The research reviewed here does not merely describe the gap; it motivates the need for the missing abstraction that Situational AI names. Whether the field ultimately formalizes that framing as a canonical paradigm is an open question, but the structural need it addresses is already present across the literature, distributed across disciplines, and waiting for the organizing principle that SCBCM and the Situational AI framework offer.

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